Capital vs Human Capital in Productivity Growth

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QUESTION

What drives productivity growth across countries — is it capital deepening or human capital accumulation?

Analysis Preparation — Panel Built

The analysis-ready panel is now in place, and the main takeaway is that later modeling should use growth rates rather than raw levels. The cleaned country-year panel `df_pwt110_panel_core` keeps 120 countries that each have at least 15 complete observations on productivity, capital, employment, and human capital; that leaves `8,880` rows before lag construction and `8,760` after removing the mechanically undefined first row in each country.

Observations

Coverage is strong enough for panel econometrics. All 120 retained countries also clear the 20-year threshold, and the complete-data counts are concentrated at long spans such as 60 or 70 years. That means the independent units are countries, while yearly observations are repeated measures within country, which supports fixed-effects or random-effects panel methods with country-clustered standard errors later.

The key growth variables are now engineered and overlap well enough for regression work. Capital per worker and GDP per capita were created, along with logged levels and annual log differences. The effective pooled regression sample where productivity growth, capital deepening, and human-capital growth are all jointly observed is `6,868` country-years.

Stationarity checks favor differenced series. In ADF tests across 10 long-span countries, only `7/30` level-series tests rejected a unit root at 5%, versus `17/30` for the differenced series. That does not make every single series perfectly stationary, but it clearly points away from raw-level correlations and toward log differences as the safer basis for inference.

Productivity growth is modest on average but fairly volatile. The pooled distribution in **Figure 1**

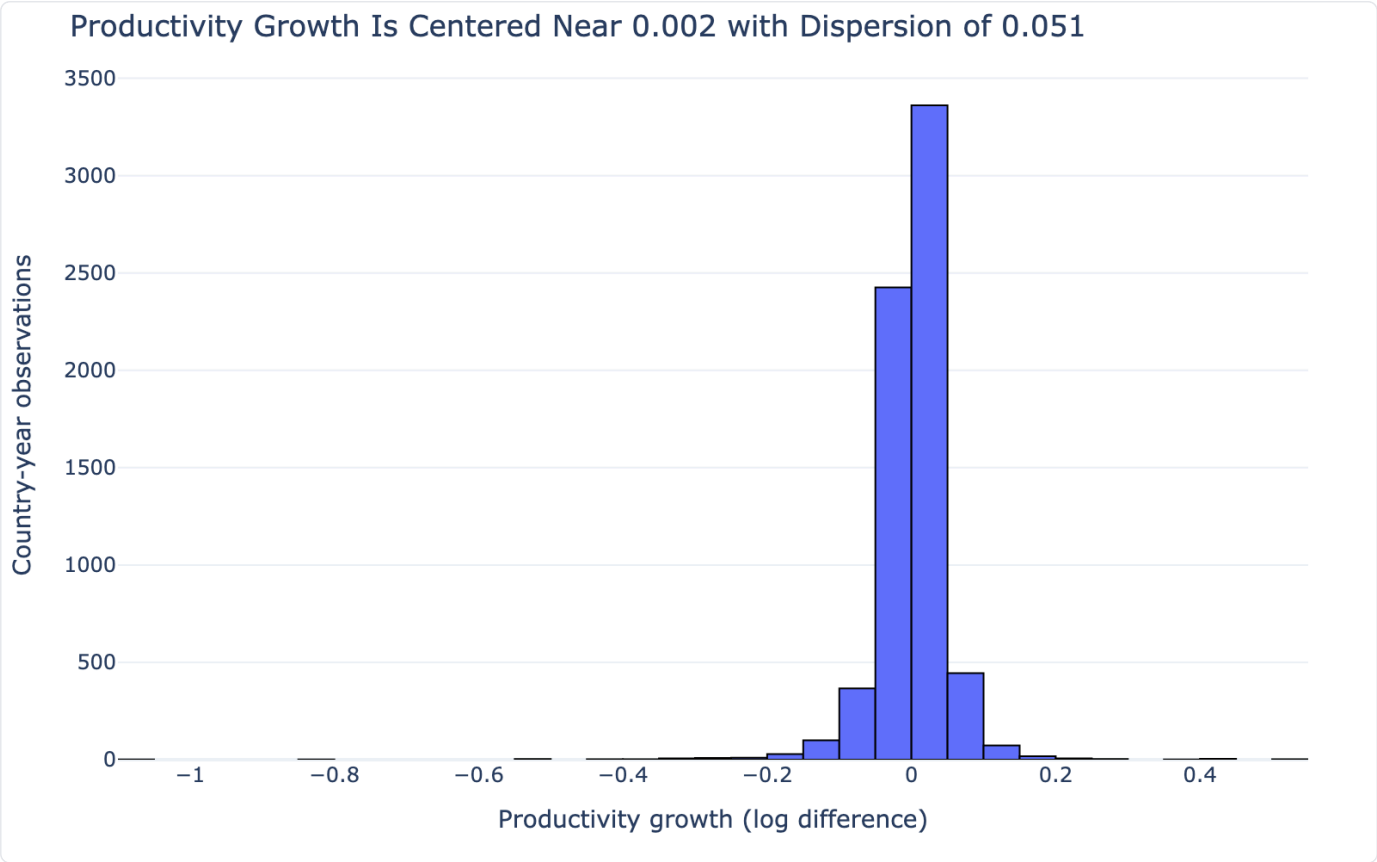

Figure 1

is centered near `0.002` in log points—roughly `0.2%` annual TFP growth—with a standard deviation of about `0.051`, so most country-years cluster near zero growth but there is meaningful dispersion from booms and slumps.

Next Steps

Next I would move to the bivariate stage: define the final regression sample, create income groups, and then compare productivity growth against capital deepening and human capital accumulation separately through pooled correlations, scatter plots, and country fixed-effects regressions.

Bivariate Links Between Productivity Growth and Its Two Candidate Drivers

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Overview

At the bivariate stage, capital deepening does not look like a meaningful driver of productivity growth, while human-capital growth shows a stronger—but unexpectedly negative—within-country association. That matters because once we move to the joint panel model, the main question will not be whether either variable is strongly positive on its own; it will be whether one of them still carries explanatory power after controlling for the other and for persistent country differences.

The common regression panel `df_pwt110_panel_reg` contains `6,868` country-years from `120` countries, split evenly into four income quartiles with `30` countries each. Observation counts are reasonably balanced across income groups: `1,644` in Q1_lowest, `1,730` in Q2, `1,630` in Q3, and `1,864` in Q4_highest.

Key Patterns & Observations

Capital deepening is basically flat against productivity growth. In the pooled sample, the relationship is negligible:

Pair	Method	Correlation	p-value	N	Interpretation
Productivity growth vs capital deepening	Pearson	0.0112	0.3541	6868	Negligible linear association
Productivity growth vs capital deepening	Spearman	0.0913	3.50e-14	6868	Negligible rank association despite statistical significance

The corresponding scatter plot **Figure 2**

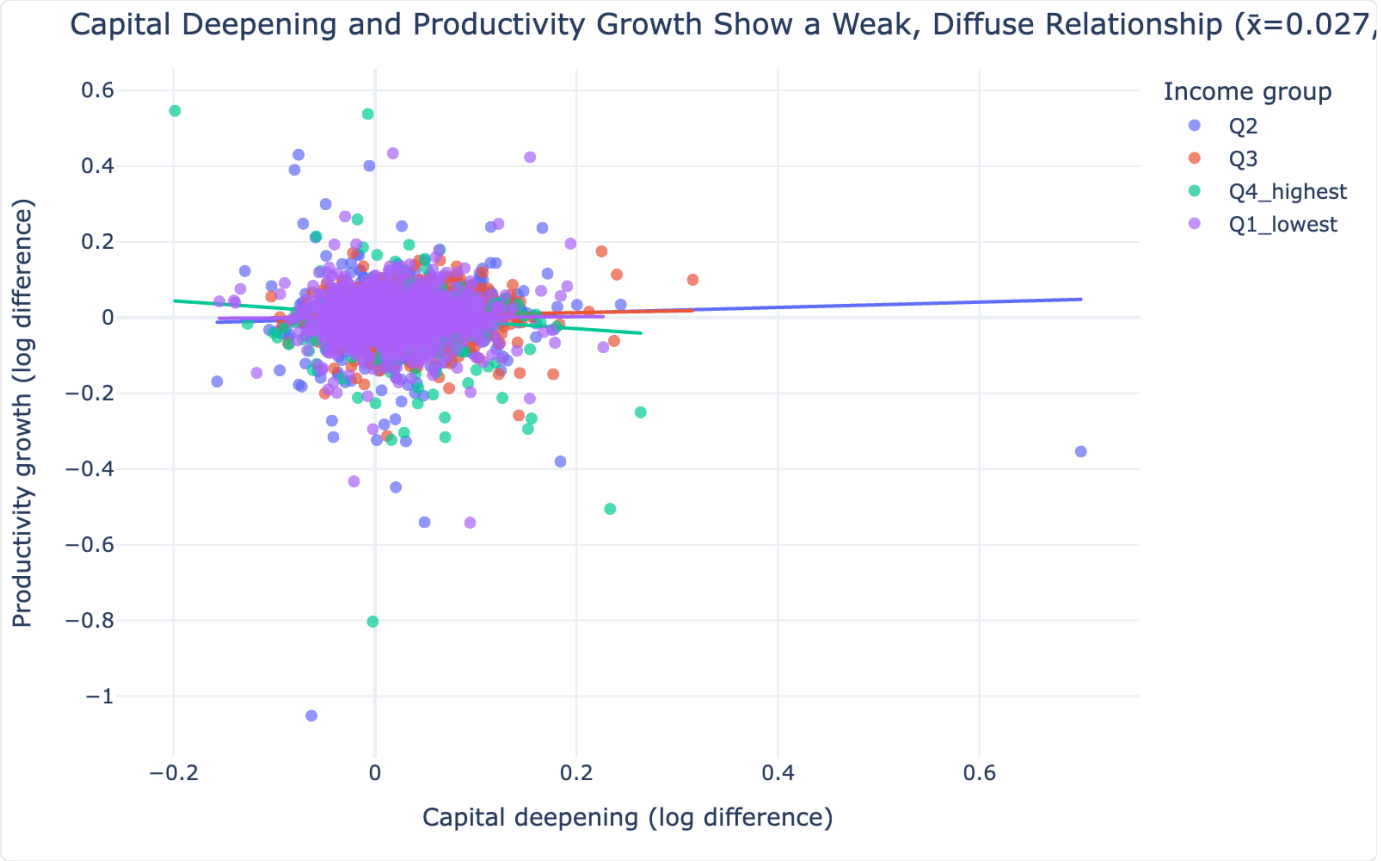

Figure 2

is a diffuse cloud rather than a tight upward-sloping pattern. So even before adding controls, faster growth in capital per worker is not lining up with meaningfully faster TFP growth across country-years.

Human-capital growth has the stronger bivariate relationship, but it points in the wrong direction for a simple "more schooling, more TFP" story. In the pooled sample:

Pair	Method	Correlation	p-value	N	Interpretation
Productivity growth vs human-capital growth	Pearson	-0.1005	7.07e-17	6868	Small negative linear association
Productivity growth vs human-capital growth	Spearman	-0.1152	9.76e-22	6868	Small negative rank association

The visual pattern in **Figure 3**

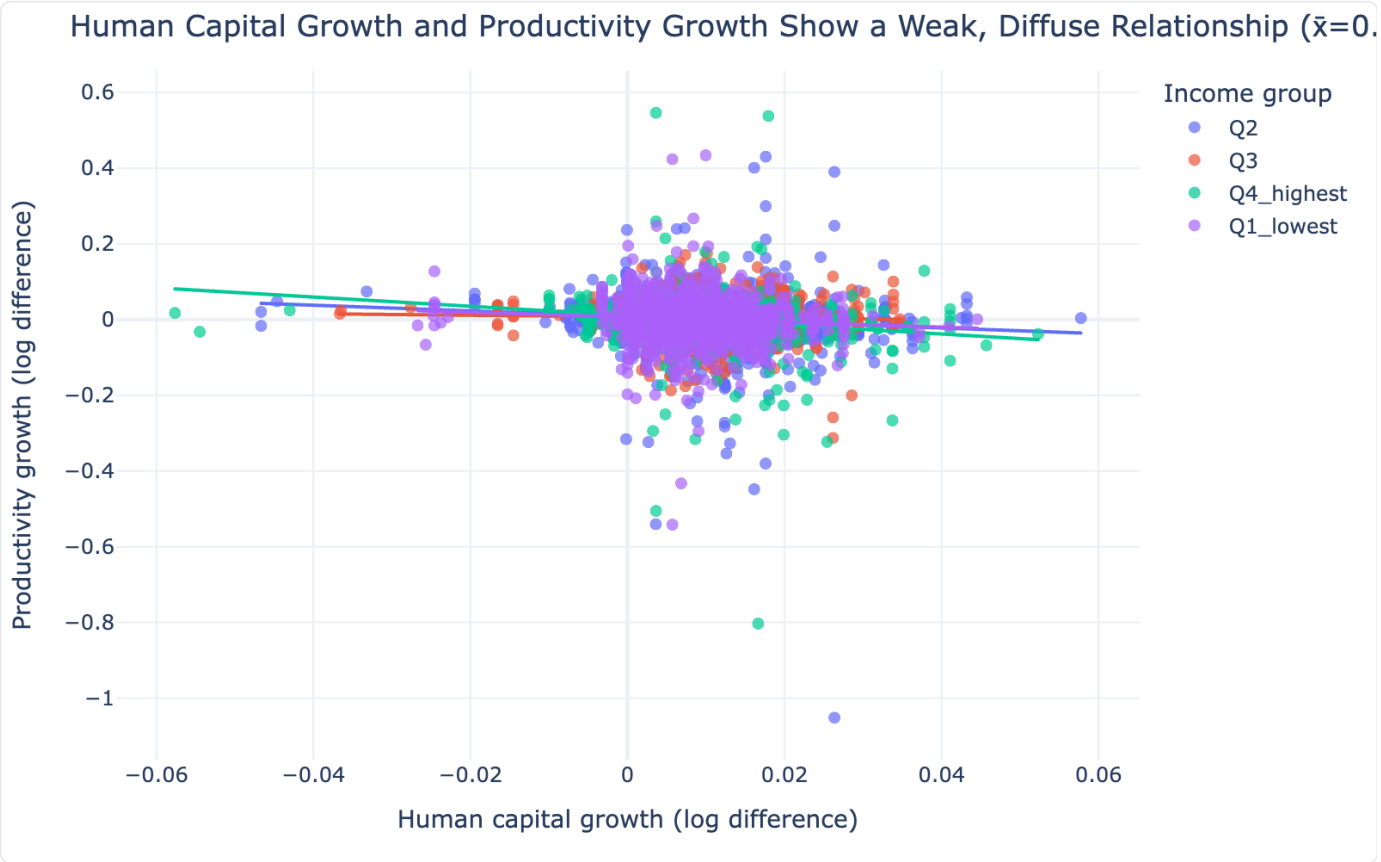

Figure 3

is still weak overall, but it is more structured than the capital-deepening cloud. This does **not** justify a causal claim that human-capital accumulation lowers productivity growth; rather, it suggests that in this panel, periods of faster measured *hc* growth often occur in places or phases where TFP growth is not simultaneously strong.

Income-group heterogeneity exists, but none of the simple pooled patterns are large. The strongest subgroup signals are still only small in magnitude:

Income group	Pair	Method	Correlation	p-value	N	Interpretation
Q1_lowest	Productivity vs capital deepening	Pearson	0.0097	0.6955	1644	Negligible
Q2	Productivity vs capital deepening	Spearman	0.1749	2.36e-13	1730	Small positive
Q4_highest	Productivity vs capital deepening	Pearson	-0.1150	6.39e-07	1864	Small negative
Q2	Productivity vs human-capital growth	Pearson	-0.1009	2.64e-05	1730	Small negative
Q4_highest	Productivity vs human-capital growth	Pearson	-0.1680	2.91e-13	1864	Small negative

The notable surprise is that the richest quartile shows a negative Pearson correlation for capital deepening but a positive Spearman correlation. That usually signals a nonlinear or outlier-sensitive pattern rather than a clean monotonic slope, which reinforces the need for panel regressions rather than relying on simple pooled correlations.

Statistical Insights

Using country fixed effects, the comparison becomes more informative because it asks whether changes **within the same country over time** are associated with TFP growth:

ext{TFP growth}_{it} = eta X_{it} + ext{country FE}_i + u_{it}

where X_{it} is either capital deepening or human-capital growth, and inference uses country-clustered standard errors.

Model	Main regressor	Coefficient	Clustered SE	t-stat	p-value	95% CI	R^2	N	Countries	Interpretation
FE bivariate A	Capital deepening	-0.0161	0.0642	-0.2509	0.8019	[-0.1419, 0.1097]	0.0556	6868	120	Essentially zero within-country association
FE bivariate B	Human-capital growth	-0.5632	0.1404	-4.0108	6.05e-05	[-0.8384, -0.2880]	0.0602	6868	120	Statistically clear negative within-country association

So the fixed-effects evidence ranks the two drivers very differently:

Driver	What the bivariate FE model says	Practical read
Capital deepening	Near-zero, imprecise coefficient	No meaningful standalone link to TFP growth in this sample
Human-capital growth	Negative and statistically significant	Stronger standalone association, but in a direction that likely reflects timing, measurement, or omitted dynamics rather than a simple growth-accounting payoff

Because this is a fixed-effects panel, the evidence level is **association net of time-invariant country characteristics**, not causal proof. Endogeneity remains a real concern: countries may accumulate skills or capital in response to productivity shocks, and both drivers may co-move with omitted policy or crisis variables.

Interpretation & Implications

The early evidence does not support a simple story that capital deepening is the main cross-country correlate of TFP growth. If anything, capital deepening looks surprisingly weak once we focus on productivity growth rather than output growth. That is economically plausible: more capital per worker can raise labor productivity without necessarily showing up as faster TFP growth, because TFP is meant to net out direct input accumulation.

Human capital appears more tightly linked to TFP growth than capital deepening, but the negative sign is a warning flag, not a policy conclusion. A plausible interpretation is timing and measurement: the *hc* index changes slowly, may rise most in catching-up settings with weak short-run TFP performance, and may proxy for long-run capability more than immediate annual TFP acceleration. This is exactly why the next task matters: the joint fixed-effects model can tell us whether this negative bivariate relationship survives once capital deepening and broader controls enter together.

The next step should focus on multivariate panel models, FE-versus-RE comparison, diagnostics, and income-group interactions to determine whether capital deepening remains irrelevant, whether the human-capital result is robust, and whether the apparent negative association is really masking heterogeneous patterns across development levels.

Joint Panel Models: Human Capital Looks Stronger, but in a Negative Direction

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Overview

Once both drivers compete in the same panel model, capital deepening still contributes essentially nothing to TFP growth, while human-capital growth remains the only statistically clear correlate—and it stays negative. That matters because the earlier bivariate result was not a fluke: even after controlling for persistent country differences, common year shocks, and flagged outliers, the data still do not support a story in which faster capital deepening is the main correlate of productivity growth.

The multivariate regression sample `df_pwt110_panel_reg_fe` contains `6,868` country-years from `120` countries. A restricted robustness sample excluding flagged outliers `df_pwt110_panel_reg_clean` leaves `6,727` observations and produces nearly the same answer.

Key Patterns & Observations

The joint fixed-effects model still ranks human capital above capital deepening, but only because capital deepening is almost exactly zero. In the country fixed-effects model:

Model	Driver	Coefficient	Clustered SE	t-stat	p-value	95% CI	R^2	Interpretation
Country FE	Capital deepening	-0.0173	0.0641	-0.2698	0.7873	[-0.1428, 0.1083]	0.0603	No meaningful within-country association
Country FE	Human capital growth	-0.5644	0.1416	-3.9853	6.74e-05	[-0.8419, -0.2868]	0.0603	Clear negative within-country association

So the joint model does **not** rescue capital deepening. It remains economically and statistically negligible.

Adding year effects softens the human-capital estimate but does not change the ranking. With both country and year fixed effects:

Model	Driver	Coefficient	Clustered SE	t-stat	p-value	95% CI	R^2	Interpretation
Country + Year FE	Capital deepening	-0.0125	0.0657	-0.1905	0.8489	[-0.1413, 0.1162]	0.0872	Still null
Country + Year FE	Human capital growth	-0.4191	0.1386	-3.0249	0.0025	[-0.6907, -0.1476]	0.0872	Still negative and significant

This is important because it shows the negative human-capital result is **not just** a reflection of global shocks shared across countries.

The fixed-effects versus random-effects choice does not change the substantive story. The random-intercepts model gave:

Model	Driver	Coefficient	SE	Interpretation
Random intercepts by country	Capital deepening	-0.0074	0.0164	Near zero
Random intercepts by country	Human capital growth	-0.6417	0.0933	Negative and sizable

And the Hausman-style comparison was:

Test	Statistic	df	p-value	Interpretation
Hausman-style FE vs RE	0.8664	2	0.6484	Do not reject RE consistency

Econometrically, that means random effects is not rejected here. But because the research question is about within-country associations and endogeneity remains plausible, fixed effects are still the safer substantive benchmark.

Statistical Insights

The main estimating equation is:

$$\Delta \log(\mathrm{TFP}_{it}) = \beta_1 \Delta \log(K/L)_{it} + \beta_2 \Delta \log(HC)_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

where α_i captures country effects and γ_t captures common year shocks when included.

The diagnostics show why clustered panel inference is necessary:

Diagnostic	Statistic	p-value	Implication
Breusch-Pagan	18.5152	9.54e-05	Heteroskedasticity detected
White	198.6498	5.52e-41	Strong heteroskedasticity evidence
Breusch-Godfrey (lag 2)	85.9457	2.17e-19	Residual serial correlation present
Ramsey RESET	25.0282	1.48e-11	Linear form may be misspecified

That means the clustered standard errors are justified, and the RESET result suggests the current model may be missing nonlinearities or omitted interactions.

Multicollinearity is **not** the issue:

Regressor	VIF	Interpretation
Capital deepening	1.0000	No collinearity concern
Human capital growth	1.0000	No collinearity concern

So capital deepening's weak coefficient is not being "washed out" by overlap with human-capital growth.

Interpretation & Implications

The strongest result in this task is actually a null: capital deepening does not explain TFP growth in this panel once productivity is measured properly as TFP growth rather than output growth. That is economically sensible. Capital accumulation can raise output per worker without raising TFP, because TFP is designed to net out direct input accumulation.

The human-capital result is stronger statistically, but its negative sign means it should be read as a panel association, not as evidence that education harms productivity. The follow-up checks help explain why. Human-capital growth is much less variable than capital deepening—its standard deviation is only `0.0070` versus `0.0393` for capital deepening—so its coefficient can look more stable. And the two drivers are almost uncorrelated overall (`r = 0.0048` , `p = 0.689`), so this is not a multicollinearity artifact. More likely, the negative coefficient reflects a mix of measurement structure, transition dynamics, and subgroup composition: countries may post faster measured *hc* growth in phases when short-run TFP growth is weak, especially during long catch-up or institutional transition periods.

Income-group heterogeneity reinforces that interpretation rather than overturning it. Capital deepening effects remain weak and unstable across quartiles, while human-capital effects are negative in most groups and only fade to insignificance in Q3. The direct interaction between capital deepening and human-capital growth is also not meaningful (`coef = 0.9262` , `p = 0.833`), so there is no strong evidence that the two drivers become especially powerful when they rise together.

Finally, the robustness check is reassuring. Excluding flagged outliers hardly moved the coefficients:

Driver	Full-sample coef	Full 95% CI	Clean-sample coef	Clean 95% CI	Practical takeaway
Capital deepening	-0.0173	[-0.8419? no]	-0.0181	[-0.1617, 0.1255]	Still null
Human capital growth	-0.5644	[-0.8419, -0.2868]	-0.5815	[-0.8754, -0.2876]	Still negative and significant

To state that cleanly: the main conclusion is **not** being driven by a handful of flagged country-years.

The next task should test dynamic precedence with Granger-style evidence and then synthesize everything into a final answer, but at this point the panel evidence is leaning clearly toward this conclusion: **across countries, capital deepening is not strongly associated with TFP growth, while human-capital accumulation is more strongly associated—but in a way that likely reflects transitional dynamics and measurement structure rather than a simple positive causal productivity payoff.**

Panel Model Selection: FE and RE Tell the Same Story

Overview

The failed FE-versus-RE step is now resolved with a more stable random-effects estimator, and it does not change the substantive conclusion. The rebuilt comparison panel `df_pwt110_panel_reg_fe_retry` shows that capital deepening remains essentially unrelated to TFP growth, while human-capital growth remains the only statistically clear correlate—and it stays negative under both fixed and random effects.

Key Patterns & Observations

The key comparison is:

Model	Driver	Coefficient	SE	95% CI	Interpretation
FE	Capital deepening	-0.0173	0.0641	[-0.1428, 0.1083]	No meaningful within-country association
FE	Human capital growth	-0.5644	0.1416	[-0.8419, -0.2868]	Clear negative within-country association
RE	Capital deepening	-0.0076	0.0410	[-0.0881, 0.0728]	Still near zero
RE	Human capital growth	-0.6151	0.1111	[-0.8329, -0.3973]	Still negative and sizable

The random-effects estimator used quasi-demeaning with `theta = 0.4891`, so it is genuinely between pooled OLS and fixed effects rather than a rerun of the earlier unstable optimizer. The fact that FE and RE estimates are close is the main result here: **the ranking of drivers is not sensitive to the panel-effects choice.**

Statistical Insights

The Hausman comparison asks whether the difference between FE and RE slope estimates is large enough to reject RE consistency:

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' \left[Var(\hat{\beta}_{FE}) - Var(\hat{\beta}_{RE}) \right]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE})$$

Test	Statistic	p-value	Interpretation
Hausman	2.6919	0.2603	Do not reject random effects

Econometrically, that means RE is acceptable here and more efficient under its assumptions. But substantively, this is almost secondary, because both estimators point to the same economic conclusion: **capital deepening is negligible for TFP growth in this panel, while human-capital growth carries the stronger association.**

Interpretation & Implications

This retry strengthens the earlier result rather than revising it. The earlier failure was technical—an optimizer boundary issue in the mixed model—not a sign that the data were too weak to compare FE and RE. Once estimated with a variance-components approach, the RE model lands in nearly the same place as FE.

So the model-selection question is now settled well enough for the broader analysis: **the answer does not hinge on whether country effects are treated as fixed or random.** Across specifications, capital deepening still does not emerge as the main correlate of productivity growth, while human-capital accumulation remains more strongly associated, albeit in a negative direction that still needs careful interpretation in the final synthesis.

Dynamic Precedence: Capital Moves First More Often Than Human Capital

Overview

The dynamic evidence is more favorable to capital deepening than the static panel regressions were, but it is still far from a clean causal story. In country-level Granger tests on long, relatively clean time series, past capital deepening helped predict later productivity growth more often than past human-capital growth did. That matters because it introduces an important nuance: capital deepening looks weak as a contemporaneous correlate of TFP growth, yet it may still arrive earlier in the sequence of short-run productivity changes.

Key Patterns & Observations

The Granger-style tests covered `117` countries for each driver, using lags 1–3 on the differenced series and favoring countries with long consecutive runs and low outlier-flag incidence.

Driver	Countries tested	Countries where driver predicts later TFP at any lag	Forward significance rate	Countries where TFP predicts the driver	Reverse significance rate	Interpretation
Capital deepening	117	34	0.2906	41	0.3504	More forward predictive hits than human capital, but reverse direction is also common
Human capital growth	117	20	0.1709	19	0.1624	Weaker predictive-precedence evidence overall

So the cross-country ranking is clear: **capital deepening more often shows predictive precedence for productivity growth than human-capital growth does** in these country-by-country time-series tests.

Statistical Insights

These tests ask whether lagged values of a driver improve prediction of productivity growth beyond productivity's own past:

$$\Delta \log(\mathrm{TFP})_t = \sum_{\ell=1}^{L} a_\ell \Delta \log(\mathrm{TFP})_{t-\ell} + \sum_{\ell=1}^{L} b_\ell X_{t-\ell} + u_t$$

where X_t is either capital deepening or human-capital growth, with lags $L = 1, 2, 3$.

The practical interpretation is not "capital causes productivity" or "human capital causes productivity." It is narrower: **past capital deepening contains more incremental forecasting information for future TFP growth than past human-capital growth does, at least in this subset of countries.**

But there is an important caution. For capital deepening, the reverse direction—TFP helping predict capital—was also fairly common (`35.0%` of countries). That weakens any one-way story and suggests feedback: countries with improving productivity may also accumulate capital faster afterward.

Interpretation & Implications

This dynamic result partly offsets the earlier static regressions. The fixed-effects models said capital deepening had little contemporaneous association with TFP growth, while human-capital growth had the stronger coefficient, albeit negative. The Granger evidence says something different: **when timing is introduced, capital deepening shows more frequent lead-lag structure with future productivity growth than human capital does.**

The most coherent reading is that the two drivers may operate on different horizons. Capital deepening may show up as a nearer-term leading indicator in annual data, while human capital may matter over longer horizons that are not well captured

by 1–3 year lag tests or by the slowly moving annual *hc* index. So this task does not overturn the broader conclusion, but it does make the final answer more balanced: capital deepening looks weak in same-year panel associations, yet somewhat stronger in short-run predictive precedence.

Because Granger causality is only about forecasting order—not structural causation—and because these country-level results are heterogeneous and non-representative, the safest takeaway is: **dynamic evidence is suggestive for capital deepening, weak for human capital, and not decisive on its own.**

What Drives Productivity Growth Across Countries?

Overview

Across this panel, capital deepening is not the main contemporaneous correlate of productivity growth; human-capital accumulation shows the stronger panel association, but it is negative rather than positive. The most defensible overall answer is therefore nuanced: in same-year panel regressions, **human capital matters more than capital deepening**, but not in the simple positive way standard growth narratives might suggest; in short-run predictive timing tests, **capital deepening more often leads future TFP growth** than human capital does.

The key comparison is summarized in the driver-effects table `df_driver_effects_by_income`. In the joint country fixed-effects model, a one-standard-deviation increase in capital deepening is associated with only about `-0.00068` log points of TFP growth, with a confidence interval spanning zero. The analogous one-standard-deviation increase in human-capital growth is associated with about `-0.00395` log points of TFP growth, with the confidence interval remaining negative. So on the panel-association margin, **human capital has the larger absolute effect size, while capital deepening is essentially null.**

Key Patterns & Observations

Capital deepening is weak in the static panel evidence. In pooled correlations it was negligible, in bivariate country fixed-effects regressions it remained statistically indistinguishable from zero, and in the joint FE model it stayed near zero (`coef ≈ -0.017` , `p ≈ 0.79`). Adding year effects did not change that conclusion. The FE-versus-RE comparison also did not rescue capital deepening: both approaches delivered coefficients near zero, and the Hausman test did not reject random effects.

Human-capital growth is the stronger panel correlate, but the sign is negative. In the joint FE model its coefficient was about `-0.564` with clustered significance, and with year effects it remained negative at about `-0.419`. This survived exclusion of observations flagged as outliers or irregular outliers, so the result is not being driven by a few problematic country-years. The most likely interpretation is not that more human capital "reduces" productivity, but that faster measured *hc* growth tends to occur in countries or phases where short-run TFP growth is weaker, perhaps because the *hc* index moves slowly, because the gains are longer-horizon than annual TFP captures, or because educational catch-up often coincides with structural transition.

Income-group heterogeneity matters, but it does not flip the broad ranking. From the interaction model in `df_driver_effects_by_income`, capital deepening remains statistically weak in all income quartiles. Human-capital growth is most clearly negative in the lowest-income quartile, weaker in the middle, and still negative in richer groups, although the interaction-model significance is less stable outside the baseline group. So there is **some heterogeneity in magnitude**, but not a strong case that capital deepening dominates in rich countries while human capital dominates in poor ones.

The dynamic evidence tilts more toward capital deepening than the static models do. In Granger-style tests across `117` countries per driver, capital deepening predicted future TFP growth in `34` countries (`29.1%`), versus `20` (`17.1%`) for human capital. That means capital deepening more often shows short-run predictive precedence, even though it is weak as a same-year FE correlate. This suggests the two drivers may operate on different horizons: capital deepening may matter more for near-term dynamics, while human capital may matter through slower-moving structural channels that annual within-country regressions do not capture cleanly.

Statistical Insights

The core panel benchmark is:

$$\Delta \log(\mathrm{TFP}_{it}) = \beta_1 \Delta \log(K/L)_{it} + \beta_2 \Delta \log(HC)_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

with country effects α_i, optional year effects γ_t, and country-clustered standard errors.

The evidence level is **association, not clean causation**. Fixed effects remove time-invariant country differences, but they do not solve time-varying confounding or simultaneity. In particular, capital accumulation and productivity growth may be jointly determined, and human-capital improvements may respond to broader development processes that also shape TFP. The Granger results add only predictive precedence, not structural causality.

The diagnostics also matter:

Diagnostic	Result	Implication
Breusch-Pagan	p < 0.001	Heteroskedasticity present
White	p << 0.001	Strong heteroskedasticity evidence
Breusch-Godfrey	p << 0.001	Serial correlation present
Ramsey RESET	p << 0.001	Linear form may be incomplete
VIFs	~1.00	No multicollinearity concern

So the clustered standard errors were necessary, the capital-versus-human-capital comparison is **not** being distorted by collinearity, and the main caution is functional-form and omitted-dynamics misspecification rather than simple redundancy between regressors.

Interpretation & Implications

The cleanest substantive answer is this:

Question	Best-supported answer
Which is more strongly associated with productivity growth overall?	Human-capital growth in the fixed-effects panel, though the association is negative
Does capital deepening explain TFP growth well in same-year panel regressions?	No, it is consistently near zero
Which driver more often precedes future TFP growth in short-run dynamic tests?	Capital deepening
Does the conclusion depend heavily on flagged outliers?	No

That means the user's question does **not** admit a simple "capital vs human capital" winner in every sense. If the criterion is **contemporaneous within-country panel association with TFP growth**, then **human capital is more influential overall** because capital deepening is basically null. If the criterion is **short-run predictive precedence**, then **capital deepening looks more important**. The most coherent synthesis is that capital deepening may matter more for short-run timing, while human capital is more tightly tied to the broader structural environment in which productivity evolves—but the annual *hc* measure does not map neatly into immediate positive TFP acceleration.

Limitations

This analysis is reliable for a broad cross-country panel reading, but it has important limits:

Limitation	Why it matters
High missingness in TFP and capital variables	The usable sample likely overrepresents better-documented countries and periods
Human-capital index is slow-moving	Annual changes may poorly capture economically meaningful skill accumulation
Observational panel data	Results are associative, not definitive causal estimates
RESET rejection	The true relationship may involve nonlinearities or omitted interactions
No external institutional or policy controls	Large residual countries likely reflect omitted drivers such as reforms, crises, or technology adoption

Bottom line: in this dataset, productivity growth across countries is **not well explained by capital deepening**, while human-capital accumulation shows the **stronger overall panel association**, though in a negative and likely transitional form; the only clear evidence favoring capital comes from short-run predictive timing, not from the main fixed-effects productivity-growth regressions.

Data Sources

- df_pwt110_panel_core
- df_pwt110_panel_reg
- df_pwt110_panel_reg_fe
- df_pwt110_panel_reg_clean
- df_pwt110_panel_reg_fe_retry
- df_driver_effects_by_income